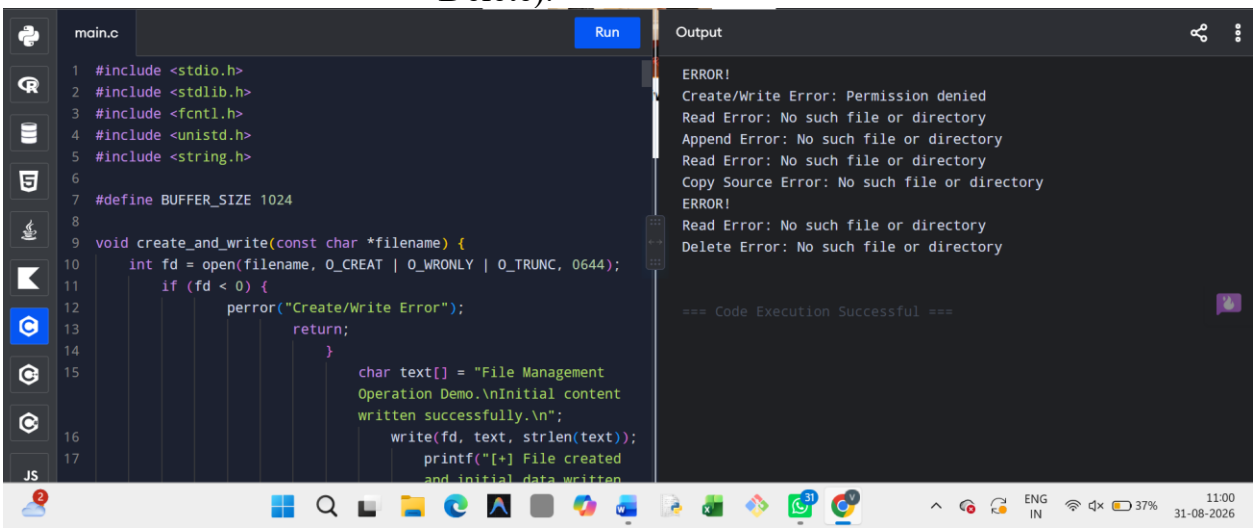
# EXPERIMENT 36:

36. Construct a C program to implement the file management operations (Create, Write, Read, Append, Copy, and Delete).




# PSEUDO CODE :

- Create & Write: Open file source.txt using open() with flags O_CREAT | O_WRONLY | O_TRUNC. Write initial text data to it using write().
- Read: Open source.txt with O_RDONLY. Read contents into a buffer using read() and display to standard output.
- Append: Open source.txt with O_WRONLY | O_APPEND. Write additional text data to the end of the file.
- Copy: Open source.txt for reading and destination.txt for writing (O_CREAT | O_WRONLY | O_TRUNC). Read chunks of data from source.txt and write them into destination.txt.
- Delete: Remove destination.txt from the filesystem using unlink() or remove()

# CODE:

```
#include <stdio.h>

#include <stdlib.h>

#include <fcntl.h>
```

```
#include <unistd.h>

#include <string.h>



#define BUFFER_SIZE
1024



void
create_and_write(const
char *filename) {

    int fd =
open(filename,
O_CREAT |
O_WRONLY |
O_TRUNC, 0644);

    if (fd < 0) {


perror("Create/Write
Error");

        return;

    }
```

```
    char text[] = "File
Management Operation
Demo.\nInitial content
written successfully.\n";

    write(fd, text,
strlen(text));

    printf("[+] File
created and initial data
written to '%s'.\n",
filename);

    close(fd);

}



void read_file(const
char *filename) {

    int fd =
open(filename,
O_RDONLY);

    if (fd < 0) {

        perror("Read
Error");
```

```
        return;
    }
    char
buffer[BUFFER_SIZE];
    int bytesRead;
    printf("\n--- Reading
Content of '%s' ---\n",
filename);
    while ((bytesRead =
read(fd, buffer,
BUFFER_SIZE - 1)) >
0) {
        buffer[bytesRead]
= '\0';
        printf("%s",
buffer);
    }
    printf("----------------
------------------\n");
    close(fd);
```

```
}

void append_file(const
char *filename) {
    int fd =
open(filename,
O_WRONLY |
O_APPEND);
    if (fd < 0) {
        perror("Append
Error");
        return;
    }
    char append_text[] =
"Appended line: Adding
new data to existing
file.\n";
    write(fd, append_text,
strlen(append_text));
    printf("[+] Appended
```

```
data to '%s'.\n",
filename);
    close(fd);
}

void copy_file(const
char *src_file, const
char *dest_file) {
    int src_fd =
open(src_file,
O_RDONLY);
    if (src_fd < 0) {
        perror("Copy
Source Error");
        return;
    }
    int dest_fd =
open(dest_file,
O_CREAT |
O_WRONLY |
```

```
O_TRUNC, 0644);
    if (dest_fd < 0) {
        perror("Copy Dest
Error");
        close(src_fd);
        return;
    }

    char
buffer[BUFFER_SIZE];
    int bytesRead;
    while ((bytesRead =
read(src_fd, buffer,
BUFFER_SIZE)) > 0) {
        write(dest_fd,
buffer, bytesRead);
    }
    printf("[+] Copied
contents from '%s' to
```

```
'%s'.\n", src_file,
dest_file);
    close(src_fd);
    close(dest_fd);
}

void delete_file(const
char *filename) {
    if (unlink(filename)
== 0) {
        printf("[+] File '%s'
successfully deleted.\n",
filename);
    } else {
        perror("Delete
Error");
    }
}
```

```
int main() {
    const char *file1 =
"demo_source.txt";
    const char *file2 =
"demo_copy.txt";

    // Perform file
operations sequentially

create_and_write(file1);
    read_file(file1);

    append_file(file1);
    read_file(file1);

    copy_file(file1, file2);
    read_file(file2);

    delete_file(file2);
```

```
    return 0;
}
```

**INPUT :**

(No direct user input required; file paths are processed internally)

**OUTPUT:**

```
[+] File created and initial data written to 'demo_source.txt'.


--- Reading Content of 'demo_source.txt' ---
File Management Operation Demo.
Initial content written successfully.
------------------------------
[+] Appended data to
```

'demo_source.txt'.

--- Reading Content of
'demo_source.txt' ---

File Management
Operation Demo.

Initial content written
successfully.

Appended line: Adding
new data to existing file.

---
---

[+] Copied contents
from 'demo_source.txt'
to 'demo_copy.txt'.

--- Reading Content of
'demo_copy.txt' ---

File Management
Operation Demo.

Initial content written
successfully.

Appended line: Adding
new data to existing file.

---

---

[+] File 'demo_copy.txt'
successfully deleted.